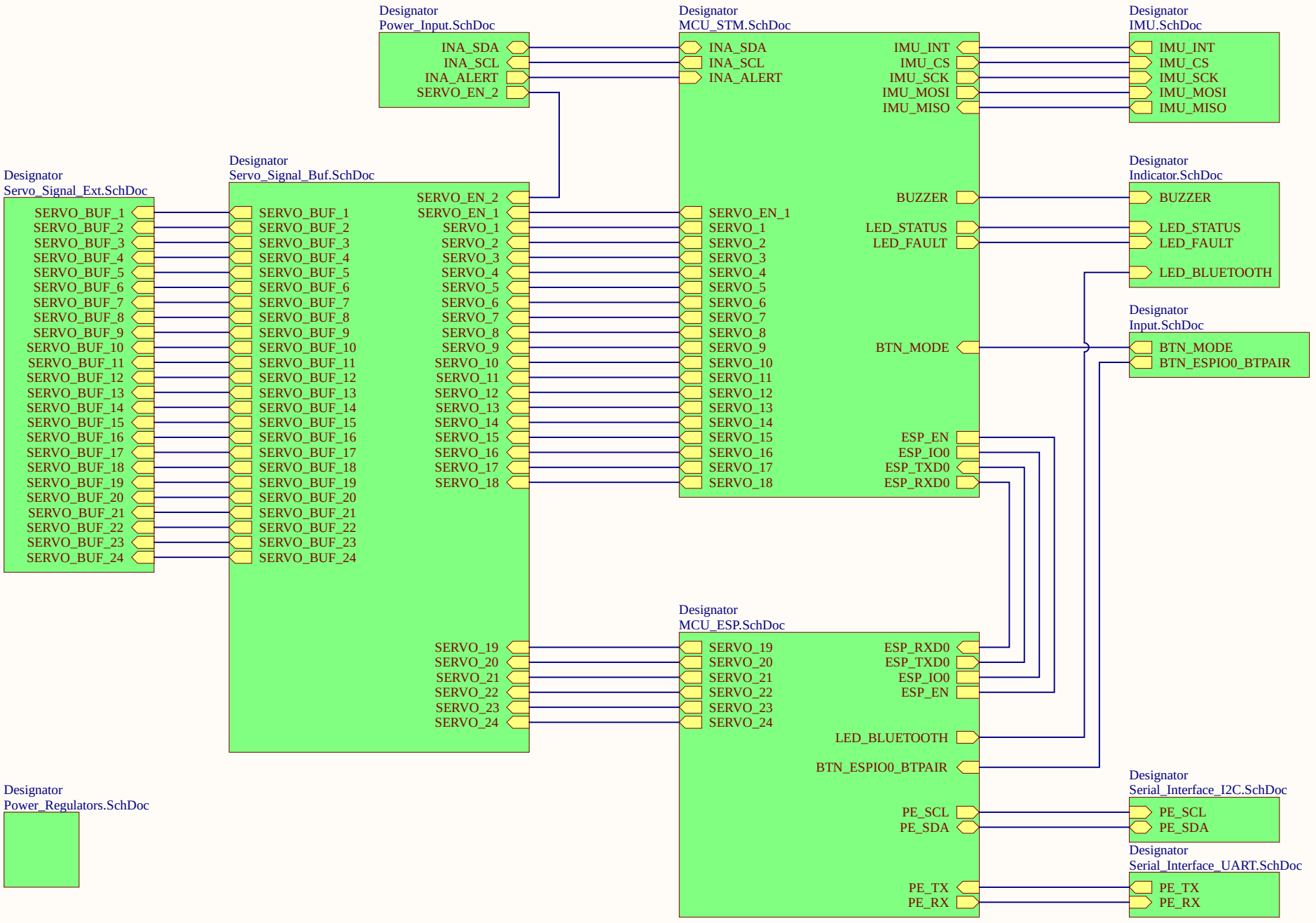
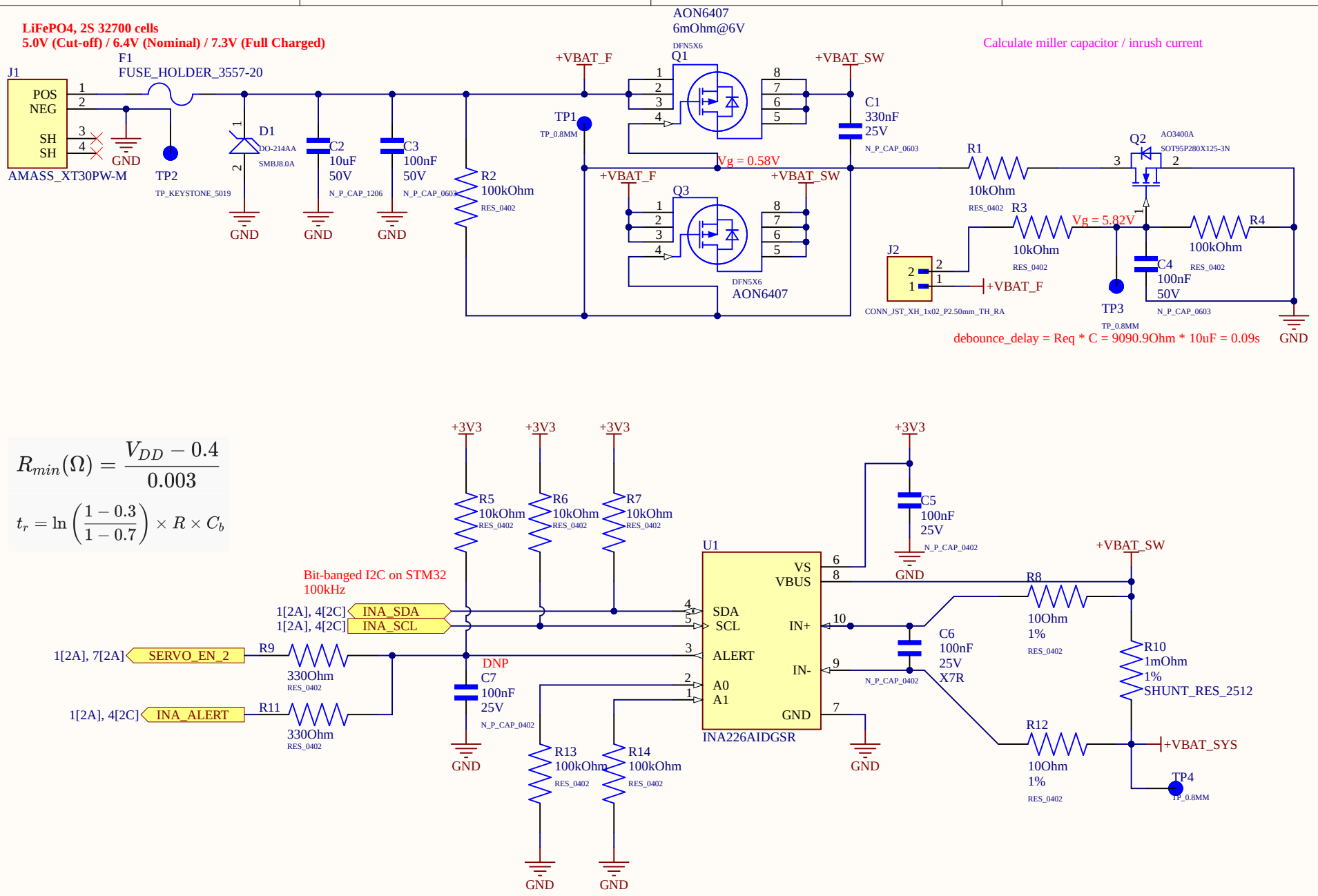
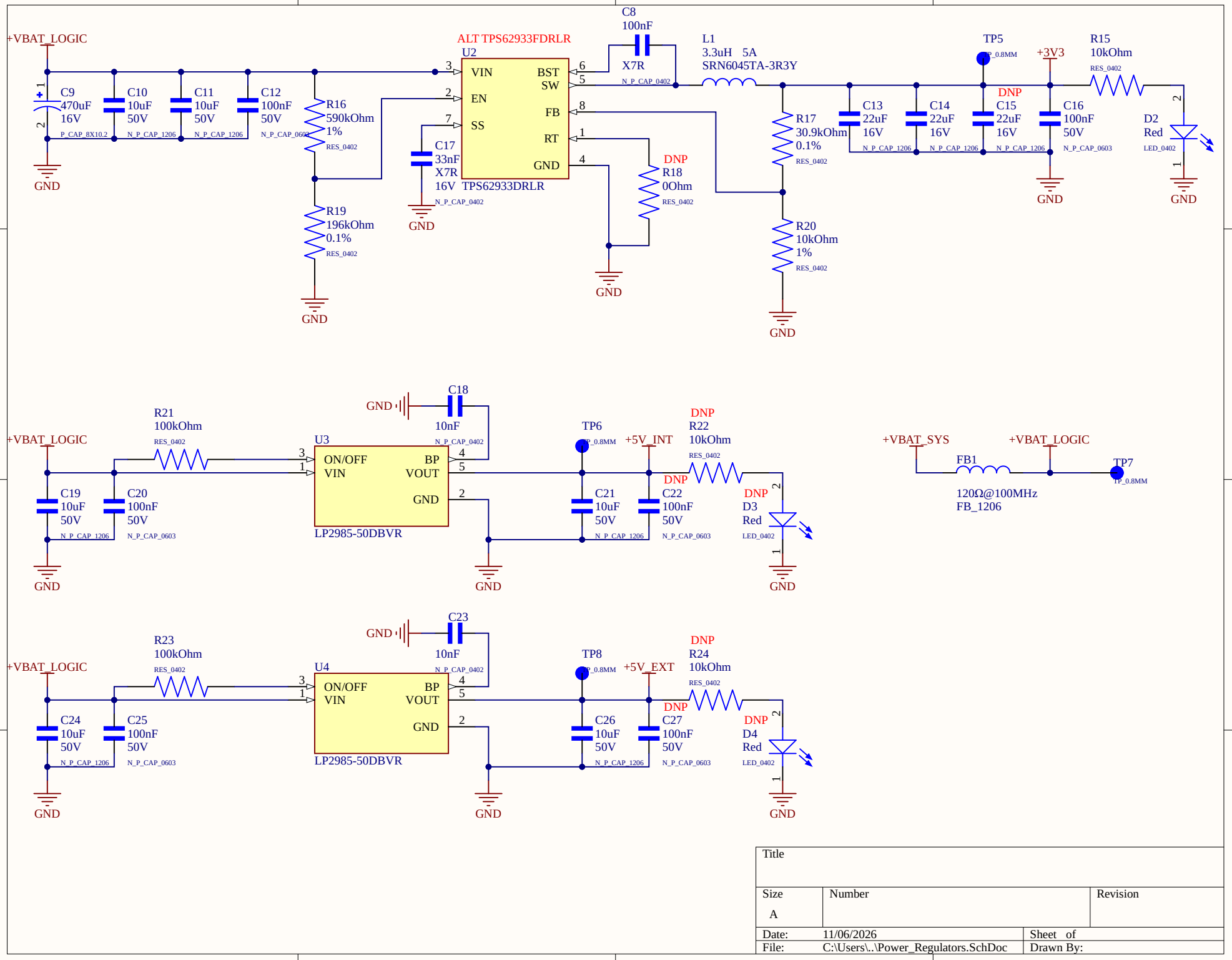
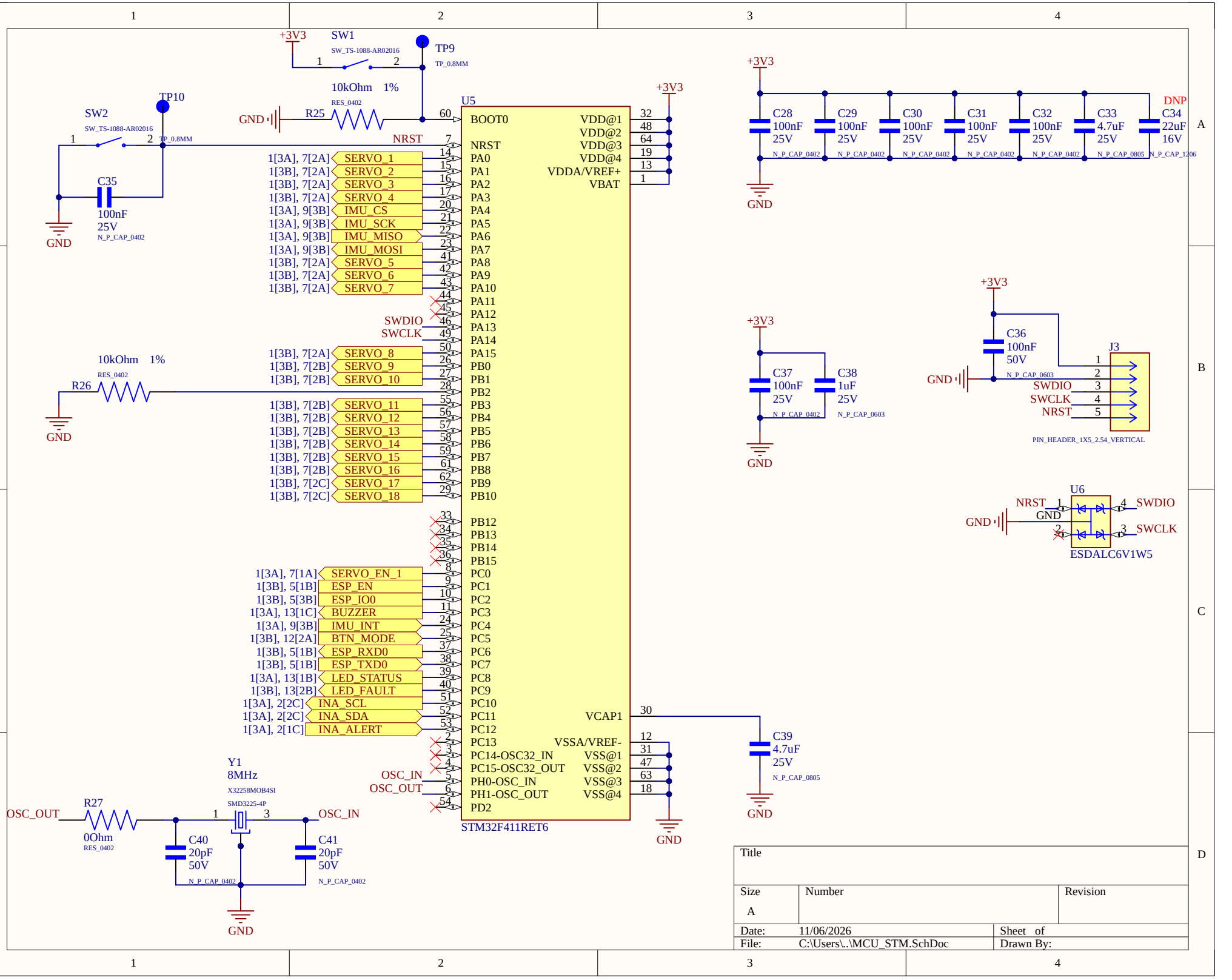
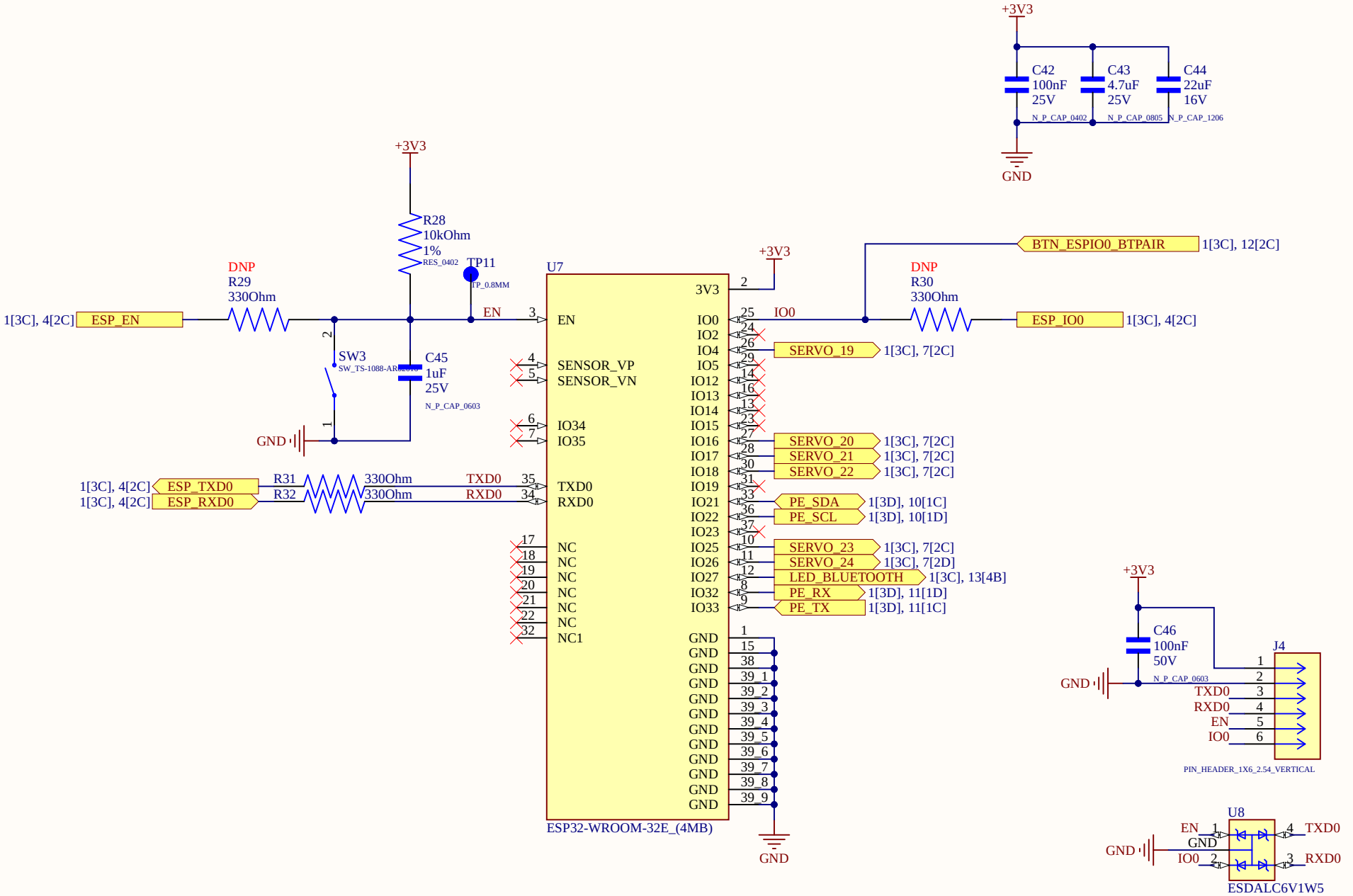

Designator
Power_Input.SchDoc
INA_SDA
INA_SCL
INA_ALERT
SERVO_EN_2
Designator
MCU_STM.SchDoc
INA_SDA
INA_SCL
INA_ALERT
IMU_INT
IMU_CS
IMU_SCK
IMU_MOSI
IMU_MISO
BUZZER
LED_STATUS
LED_FAULT
BTN_MODE
ESP_EN
ESP_IO0
ESP_TXD0
ESP_RXD0
SERVO_EN_1
SERVO_1
SERVO_2
SERVO_3
SERVO_4
SERVO_5
SERVO_6
SERVO_7
SERVO_8
SERVO_9
SERVO_10
SERVO_11
SERVO_12
SERVO_13
SERVO_14
SERVO_15
SERVO_16
SERVO_17
SERVO_18
Designator
IMU.SchDoc
IMU_INT
IMU_CS
IMU_SCK
IMU_MOSI
IMU_MISO
Designator
Indicator.SchDoc
BUZZER
LED_STATUS
LED_FAULT
LED_BLUETOOTH
Designator
Input.SchDoc
BTN_MODE
BTN_ESPIO0_BTPAIR
Designator
Servo_Signal_Ext.SchDoc
SERVO_BUF_1
SERVO_BUF_2
SERVO_BUF_3
SERVO_BUF_4
SERVO_BUF_5
SERVO_BUF_6
SERVO_BUF_7
SERVO_BUF_8
SERVO_BUF_9
SERVO_BUF_10
SERVO_BUF_11
SERVO_BUF_12
SERVO_BUF_13
SERVO_BUF_14
SERVO_BUF_15
SERVO_BUF_16
SERVO_BUF_17
SERVO_BUF_18
SERVO_BUF_19
SERVO_BUF_20
SERVO_BUF_21
SERVO_BUF_22
SERVO_BUF_23
SERVO_BUF_24
Designator
Servo_Signal_Buf.SchDoc
SERVO_EN_2
SERVO_EN_1
SERVO_1
SERVO_2
SERVO_3
SERVO_4
SERVO_5
SERVO_6
SERVO_7
SERVO_8
SERVO_9
SERVO_10
SERVO_11
SERVO_12
SERVO_13
SERVO_14
SERVO_15
SERVO_16
SERVO_17
SERVO_18
SERVO_19
SERVO_20
SERVO_21
SERVO_22
SERVO_23
SERVO_24
Designator
MCU_ESP.SchDoc
SERVO_19
SERVO_20
SERVO_21
SERVO_22
SERVO_23
SERVO_24
ESP_RXD0
ESP_TXD0
ESP_IO0
ESP_EN
LED_BLUETOOTH
BTN_ESPIO0_BTPAIR
PE_SCL
PE_SDA
PE_TX
PE_RX
Designator
Serial_Interface_I2C.SchDoc
PE_SCL
PE_SDA
Designator
Serial_Interface_UART.SchDoc
PE_TX
PE_RX
Designator
Power_Regulators.SchDoc
Designator
Mechanical.SchDoc
Designator
Servo_Power.SchDoc
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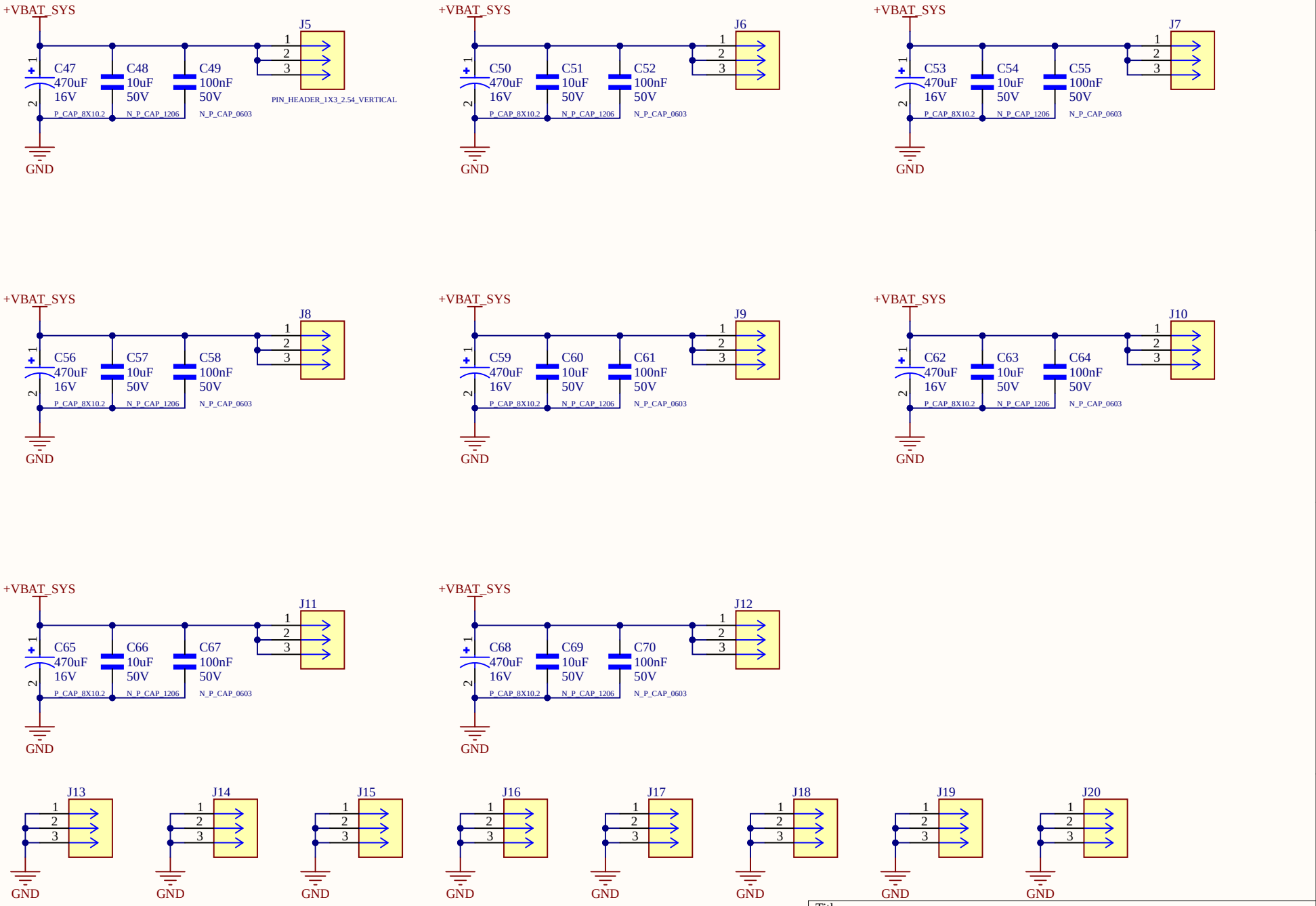

LiFePO4, 2S 32700 cells
5.0V (Cut-off) / 6.4V (Nominal) / 7.3V (Full Charged)
F1
FUSE_HOLDER_3557-20
J1
POS
NEG
SH
SH
AMASS_XT30PW-M
GND
TP2
TP_KEYSTONE_5019
D1
DO-214AA
SMBJ8.0A
C2
10uF
50V
N_P_CAP_1206
C3
100nF
50V
N_P_CAP_0603
R2
100kOhm
RES_0402
AON6407
6mOhm@6V
DFN5X6
Q1
Q3
+VBAT_F
+VBAT_SW
TP1
TP_0.8MM
Vg = 0.58V
C1
330nF
25V
N_P_CAP_0603
R1
10kOhm
RES_0402
Q2
AO3400A
SOT95P280X125-3N
R3
10kOhm
Vg = 5.82V
R4
100kOhm
C4
100nF
50V
J2
CONN_JST_XH_1x02_P2.50mm_TH_RA
TP3
Calculate miller capacitor / inrush current
debounce_delay = Req * C = 9090.9Ohm * 10uF = 0.09s
$R_{min}(\Omega) = \frac{V_{DD} - 0.4}{0.003}$
$t_r = \ln\left(\frac{1-0.3}{1-0.7}\right) \times R \times C_b$
+3V3
R5
10kOhm
R6
10kOhm
R7
10kOhm
Bit-banged I2C on STM32
100kHz
1[2A], 4[2C]
INA_SDA
INA_SCL
1[2A], 7[2A]
SERVO_EN_2
R9
330Ohm
INA_ALERT
R11
330Ohm
DNP
C7
100nF
25V
U1
VS
VBUS
SDA
SCL
ALERT
A0
A1
IN+
IN-
GND
INA226AIDGSR
C5
100nF
25V
C6
100nF
25V
X7R
R8
10Ohm
1%
R10
1mOhm
1%
SHUNT_RES_2512
R12
10Ohm
1%
+VBAT_SYS
TP4
R13
100kOhm
R14
100kOhm
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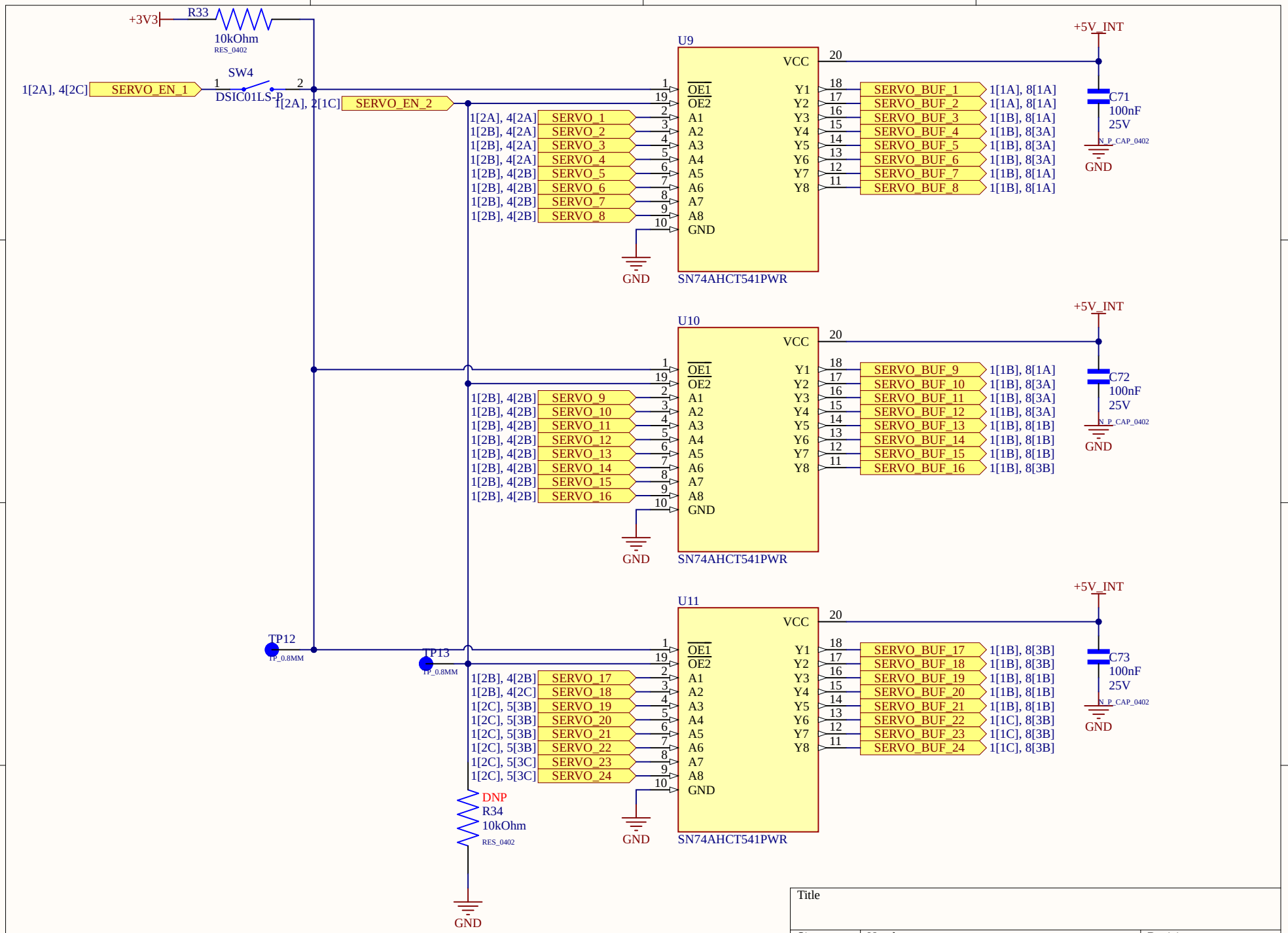

+VBAT_LOGIC
C9
470uF
16V
P_CAP_8X10.2
C10
10uF
50V
N_P_CAP_1206
C11
10uF
50V
N_P_CAP_1206
C12
100nF
50V
N_P_CAP_0603
R16
590kOhm
1%
RES_0402
R19
196kOhm
0.1%
RES_0402
GND
ALT TPS62933FDRLR
U2
VIN
EN
SS
BST
SW
FB
RT
GND
TPS62933DRLR
C17
33nF
X7R
16V
N_P_CAP_0402
C8
100nF
X7R
N_P_CAP_0402
L1
3.3uH 5A
SRN6045TA-3R3Y
DNP
R18
0Ohm
RES_0402
R17
30.9kOhm
0.1%
RES_0402
R20
10kOhm
1%
RES_0402
C13
22uF
16V
N_P_CAP_1206
C14
22uF
16V
N_P_CAP_1206
TP5
TP_0.8MM
DNP
C15
22uF
16V
N_P_CAP_1206
+3V3
C16
100nF
50V
N_P_CAP_0603
R15
10kOhm
RES_0402
D2
Red
LED_0402
R21
100kOhm
RES_0402
C19
10uF
50V
N_P_CAP_1206
C20
100nF
50V
N_P_CAP_0603
U3
ON/OFF
VIN
BP
VOUT
GND
LP2985-50DBVR
C18
10nF
N_P_CAP_0402
TP6
+5V_INT
C21
10uF
50V
N_P_CAP_1206
DNP
R22
10kOhm
RES_0402
DNP
C22
100nF
50V
N_P_CAP_0603
DNP
D3
Red
LED_0402
+VBAT_SYS
FB1
120Ω@100MHz
FB_1206
+VBAT_LOGIC
TP7
TP_0.8MM
R23
100kOhm
RES_0402
C24
10uF
50V
N_P_CAP_1206
C25
100nF
50V
N_P_CAP_0603
U4
ON/OFF
VIN
BP
VOUT
GND
LP2985-50DBVR
C23
10nF
N_P_CAP_0402
TP8
+5V_EXT
C26
10uF
50V
N_P_CAP_1206
DNP
R24
10kOhm
RES_0402
DNP
C27
100nF
50V
N_P_CAP_0603
DNP
D4
Red
LED_0402
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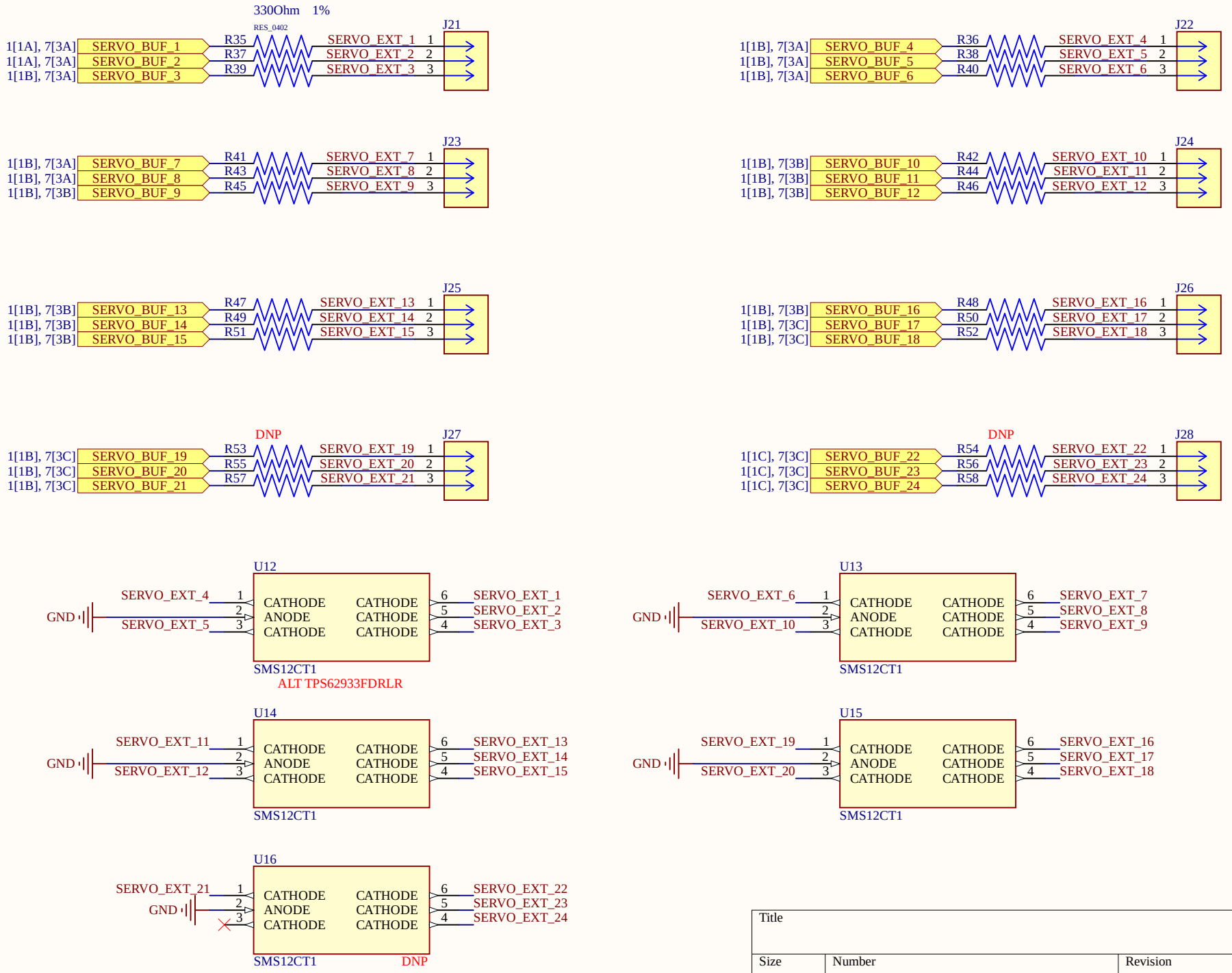

+3V3
SW1
SW_TS-1088-AR02016
TP9
TP_0.8MM
10kOhm 1%
RES_0402
R25
GND
TP10
SW2
SW_TS-1088-AR02016
TP_0.8MM
C35
100nF
25V
N_P_CAP_0402
U5
BOOT0
NRST
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
PH0-OSC_IN
PH1-OSC_OUT
PD2
VDD@1
VDD@2
VDD@3
VDD@4
VDDA/VREF+
VBAT
VCAP1
VSSA/VREF-
VSS@1
VSS@2
VSS@3
VSS@4
STM32F411RET6
SERVO_1
SERVO_2
SERVO_3
SERVO_4
IMU_CS
IMU_SCK
IMU_MISO
IMU_MOSI
SERVO_5
SERVO_6
SERVO_7
SERVO_8
SERVO_9
SERVO_10
SERVO_11
SERVO_12
SERVO_13
SERVO_14
SERVO_15
SERVO_16
SERVO_17
SERVO_18
SERVO_EN_1
ESP_EN
ESP_IO0
BUZZER
IMU_INT
BTN_MODE
ESP_RXD0
ESP_TXD0
LED_STATUS
LED_FAULT
INA_SCL
INA_SDA
INA_ALERT
SWDIO
SWCLK
OSC_IN
OSC_OUT
10kOhm 1%
RES_0402
R26
Y1
8MHz
X32258MOB4SI
SMD3225-4P
R27
0Ohm
RES_0402
C40
20pF
50V
N_P_CAP_0402
C41
20pF
50V
N_P_CAP_0402
C39
4.7uF
25V
N_P_CAP_0805
C28
100nF
25V
N_P_CAP_0402
C29
100nF
25V
N_P_CAP_0402
C30
100nF
25V
N_P_CAP_0402
C31
100nF
25V
N_P_CAP_0402
C32
100nF
25V
N_P_CAP_0402
C33
4.7uF
25V
N_P_CAP_0805
C34
22uF
16V
N_P_CAP_1206
DNP
C37
100nF
25V
N_P_CAP_0402
C38
1uF
25V
N_P_CAP_0603
C36
100nF
50V
N_P_CAP_0603
J3
PIN_HEADER_1X5_2.54_VERTICAL
U6
ESDALC6V1W5
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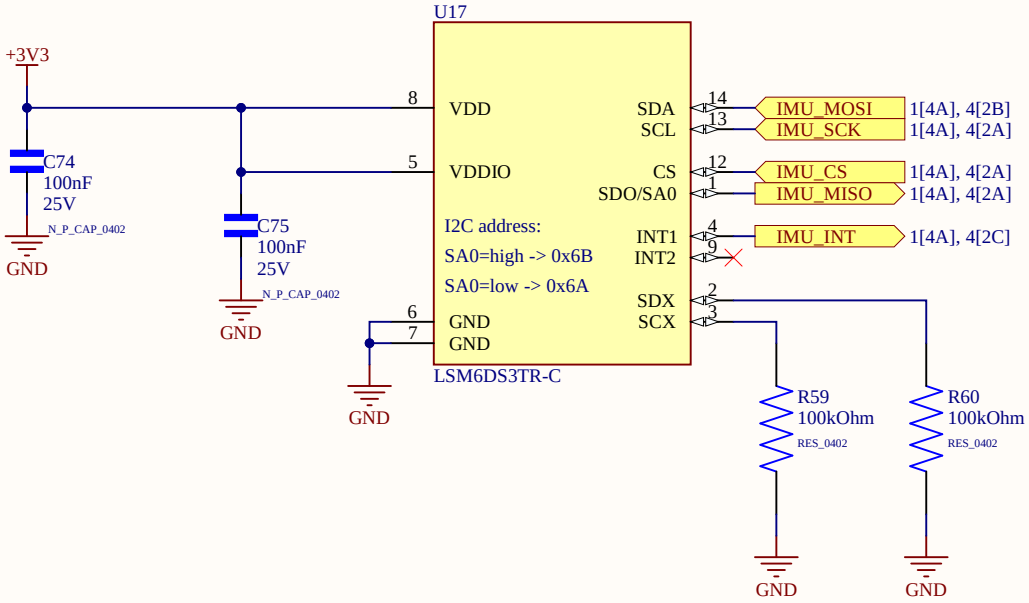


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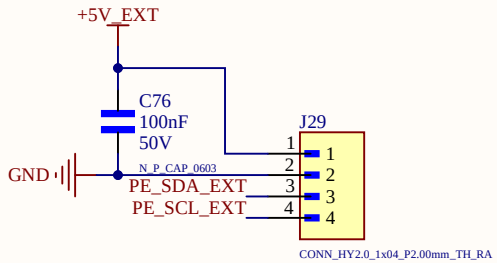

+VBAT_SYS
C47
470uF
16V
P_CAP_8X10.2
C48
10uF
50V
N_P_CAP_1206
C49
100nF
50V
N_P_CAP_0603
J5
PIN_HEADER_1X3_2.54_VERTICAL
GND
+VBAT_SYS
C50
470uF
16V
C51
10uF
50V
C52
100nF
50V
J6
GND
+VBAT_SYS
C53
470uF
16V
C54
10uF
50V
C55
100nF
50V
J7
GND
+VBAT_SYS
C56
470uF
16V
C57
10uF
50V
C58
100nF
50V
J8
GND
+VBAT_SYS
C59
470uF
16V
C60
10uF
50V
C61
100nF
50V
J9
GND
+VBAT_SYS
C62
470uF
16V
C63
10uF
50V
C64
100nF
50V
J10
GND
+VBAT_SYS
C65
470uF
16V
C66
10uF
50V
C67
100nF
50V
J11
GND
+VBAT_SYS
C68
470uF
16V
C69
10uF
50V
C70
100nF
50V
J12
GND
J13
GND
J14
GND
J15
GND
J16
GND
J17
GND
J18
GND
J19
GND
J20
GND
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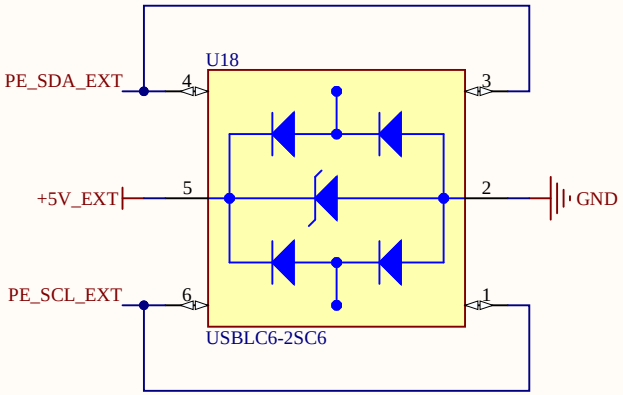

+3V3
R33
10kOhm
RES_0402
SW4
1[2A], 4[2C]
SERVO_EN_1
DSIC01LS-P
1[2A], 2[1C]
SERVO_EN_2
U9
VCC
OE1
OE2
A1
A2
A3
A4
A5
A6
A7
A8
GND
Y1
Y2
Y3
Y4
Y5
Y6
Y7
Y8
SN74AHCT541PWR
1[2A], 4[2A] SERVO_1
1[2B], 4[2A] SERVO_2
1[2B], 4[2A] SERVO_3
1[2B], 4[2A] SERVO_4
1[2B], 4[2B] SERVO_5
1[2B], 4[2B] SERVO_6
1[2B], 4[2B] SERVO_7
1[2B], 4[2B] SERVO_8
SERVO_BUF_1 1[1A], 8[1A]
SERVO_BUF_2 1[1A], 8[1A]
SERVO_BUF_3 1[1B], 8[1A]
SERVO_BUF_4 1[1B], 8[3A]
SERVO_BUF_5 1[1B], 8[3A]
SERVO_BUF_6 1[1B], 8[3A]
SERVO_BUF_7 1[1B], 8[1A]
SERVO_BUF_8 1[1B], 8[1A]
+5V_INT
C71
100nF
25V
N_P_CAP_0402
GND
U10
1[2B], 4[2B] SERVO_9
1[2B], 4[2B] SERVO_10
1[2B], 4[2B] SERVO_11
1[2B], 4[2B] SERVO_12
1[2B], 4[2B] SERVO_13
1[2B], 4[2B] SERVO_14
1[2B], 4[2B] SERVO_15
1[2B], 4[2B] SERVO_16
SERVO_BUF_9 1[1B], 8[1A]
SERVO_BUF_10 1[1B], 8[3A]
SERVO_BUF_11 1[1B], 8[3A]
SERVO_BUF_12 1[1B], 8[3A]
SERVO_BUF_13 1[1B], 8[1B]
SERVO_BUF_14 1[1B], 8[1B]
SERVO_BUF_15 1[1B], 8[1B]
SERVO_BUF_16 1[1B], 8[3B]
C72
100nF
25V
U11
TP12
TP_0.8MM
TP13
1[2B], 4[2B] SERVO_17
1[2B], 4[2C] SERVO_18
1[2C], 5[3B] SERVO_19
1[2C], 5[3B] SERVO_20
1[2C], 5[3B] SERVO_21
1[2C], 5[3B] SERVO_22
1[2C], 5[3C] SERVO_23
1[2C], 5[3C] SERVO_24
SERVO_BUF_17 1[1B], 8[3B]
SERVO_BUF_18 1[1B], 8[3B]
SERVO_BUF_19 1[1B], 8[1B]
SERVO_BUF_20 1[1B], 8[1B]
SERVO_BUF_21 1[1B], 8[1B]
SERVO_BUF_22 1[1C], 8[3B]
SERVO_BUF_23 1[1C], 8[3B]
SERVO_BUF_24 1[1C], 8[3B]
C73
100nF
25V
DNP
R34
10kOhm
RES_0402
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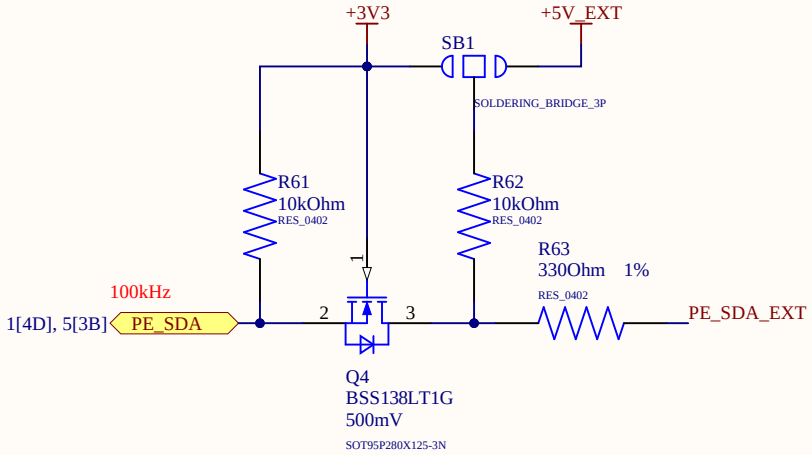

330Ohm 1%
RES_0402
J21
1[1A], 7[3A] SERVO_BUF_1 R35 SERVO_EXT_1 1
1[1A], 7[3A] SERVO_BUF_2 R37 SERVO_EXT_2 2
1[1B], 7[3A] SERVO_BUF_3 R39 SERVO_EXT_3 3
J22
1[1B], 7[3A] SERVO_BUF_4 R36 SERVO_EXT_4 1
1[1B], 7[3A] SERVO_BUF_5 R38 SERVO_EXT_5 2
1[1B], 7[3A] SERVO_BUF_6 R40 SERVO_EXT_6 3
J23
1[1B], 7[3A] SERVO_BUF_7 R41 SERVO_EXT_7 1
1[1B], 7[3A] SERVO_BUF_8 R43 SERVO_EXT_8 2
1[1B], 7[3B] SERVO_BUF_9 R45 SERVO_EXT_9 3
J24
1[1B], 7[3B] SERVO_BUF_10 R42 SERVO_EXT_10 1
1[1B], 7[3B] SERVO_BUF_11 R44 SERVO_EXT_11 2
1[1B], 7[3B] SERVO_BUF_12 R46 SERVO_EXT_12 3
J25
1[1B], 7[3B] SERVO_BUF_13 R47 SERVO_EXT_13 1
1[1B], 7[3B] SERVO_BUF_14 R49 SERVO_EXT_14 2
1[1B], 7[3B] SERVO_BUF_15 R51 SERVO_EXT_15 3
J26
1[1B], 7[3B] SERVO_BUF_16 R48 SERVO_EXT_16 1
1[1B], 7[3C] SERVO_BUF_17 R50 SERVO_EXT_17 2
1[1B], 7[3C] SERVO_BUF_18 R52 SERVO_EXT_18 3
DNP
J27
1[1B], 7[3C] SERVO_BUF_19 R53 SERVO_EXT_19 1
1[1B], 7[3C] SERVO_BUF_20 R55 SERVO_EXT_20 2
1[1B], 7[3C] SERVO_BUF_21 R57 SERVO_EXT_21 3
DNP
J28
1[1C], 7[3C] SERVO_BUF_22 R54 SERVO_EXT_22 1
1[1C], 7[3C] SERVO_BUF_23 R56 SERVO_EXT_23 2
1[1C], 7[3C] SERVO_BUF_24 R58 SERVO_EXT_24 3
U12
SERVO_EXT_4 1
GND 2
SERVO_EXT_5 3
CATHODE CATHODE
ANODE CATHODE
CATHODE CATHODE
6 SERVO_EXT_1
5 SERVO_EXT_2
4 SERVO_EXT_3
SMS12CT1
ALT TPS62933FDRLR
U13
SERVO_EXT_6 1
GND 2
SERVO_EXT_10 3
CATHODE CATHODE
ANODE CATHODE
CATHODE CATHODE
6 SERVO_EXT_7
5 SERVO_EXT_8
4 SERVO_EXT_9
SMS12CT1
U14
SERVO_EXT_11 1
GND 2
SERVO_EXT_12 3
CATHODE CATHODE
ANODE CATHODE
CATHODE CATHODE
6 SERVO_EXT_13
5 SERVO_EXT_14
4 SERVO_EXT_15
SMS12CT1
U15
SERVO_EXT_19 1
GND 2
SERVO_EXT_20 3
CATHODE CATHODE
ANODE CATHODE
CATHODE CATHODE
6 SERVO_EXT_16
5 SERVO_EXT_17
4 SERVO_EXT_18
SMS12CT1
U16
SERVO_EXT_21 1
GND 2
3
CATHODE CATHODE
ANODE CATHODE
CATHODE CATHODE
6 SERVO_EXT_22
5 SERVO_EXT_23
4 SERVO_EXT_24
SMS12CT1 DNP
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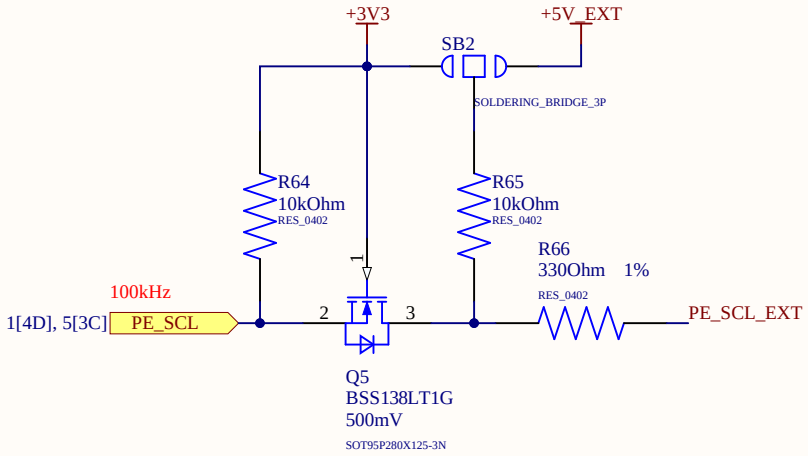


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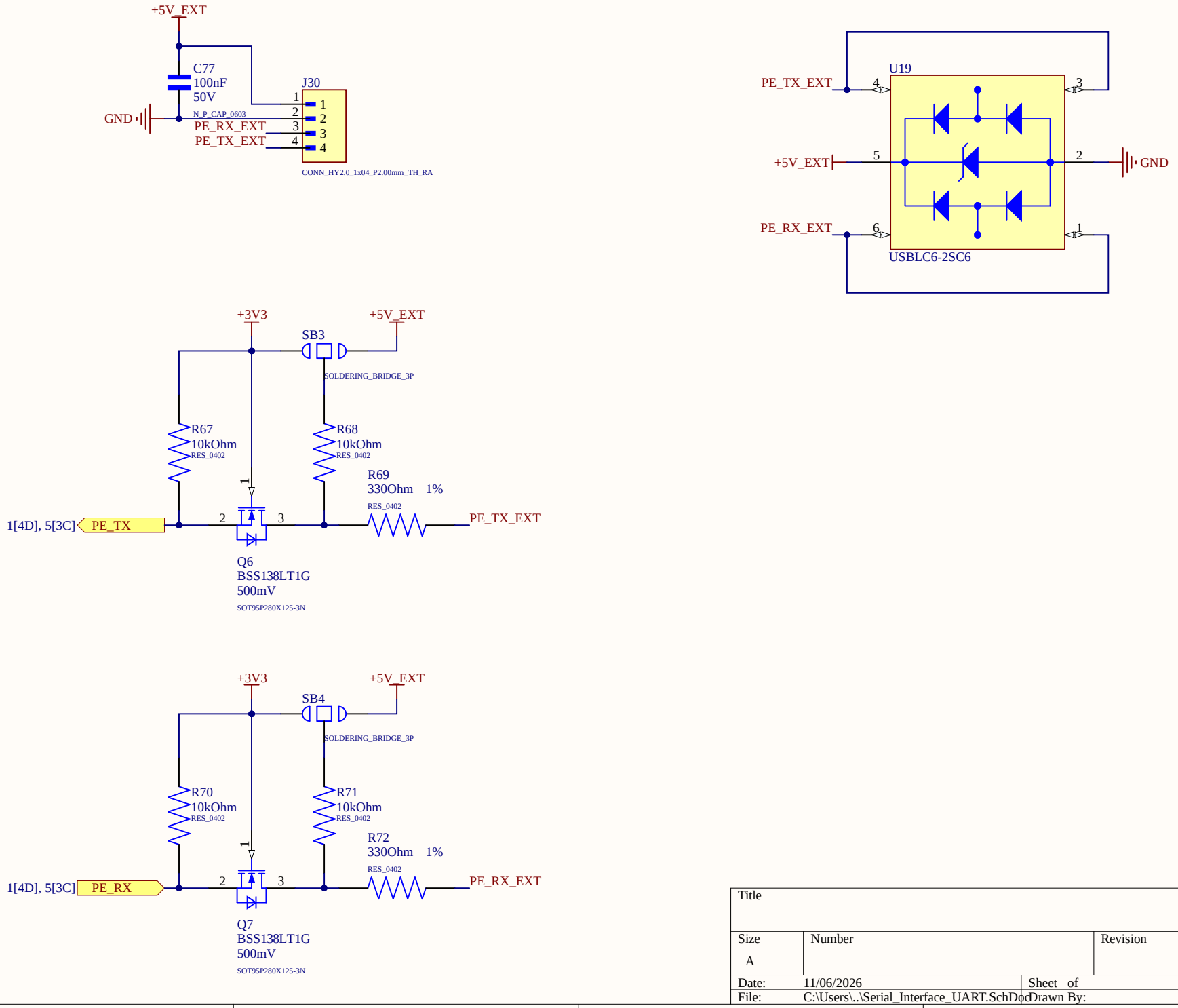


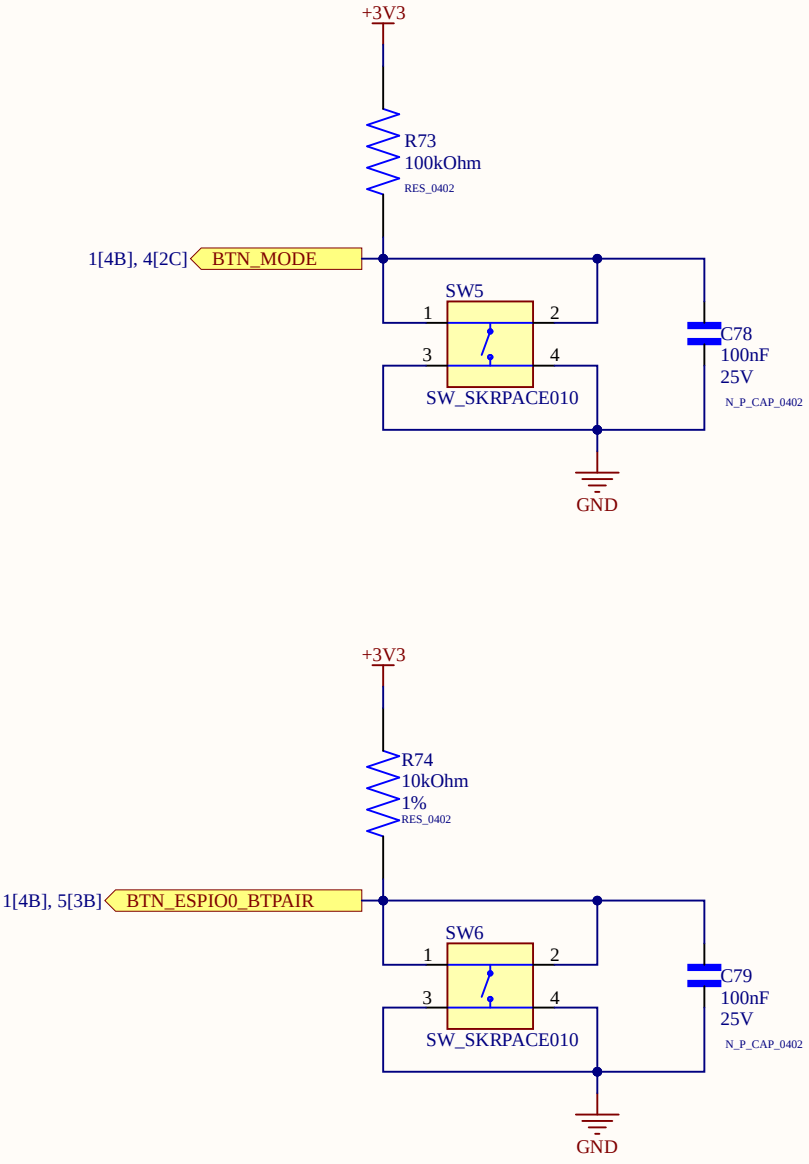




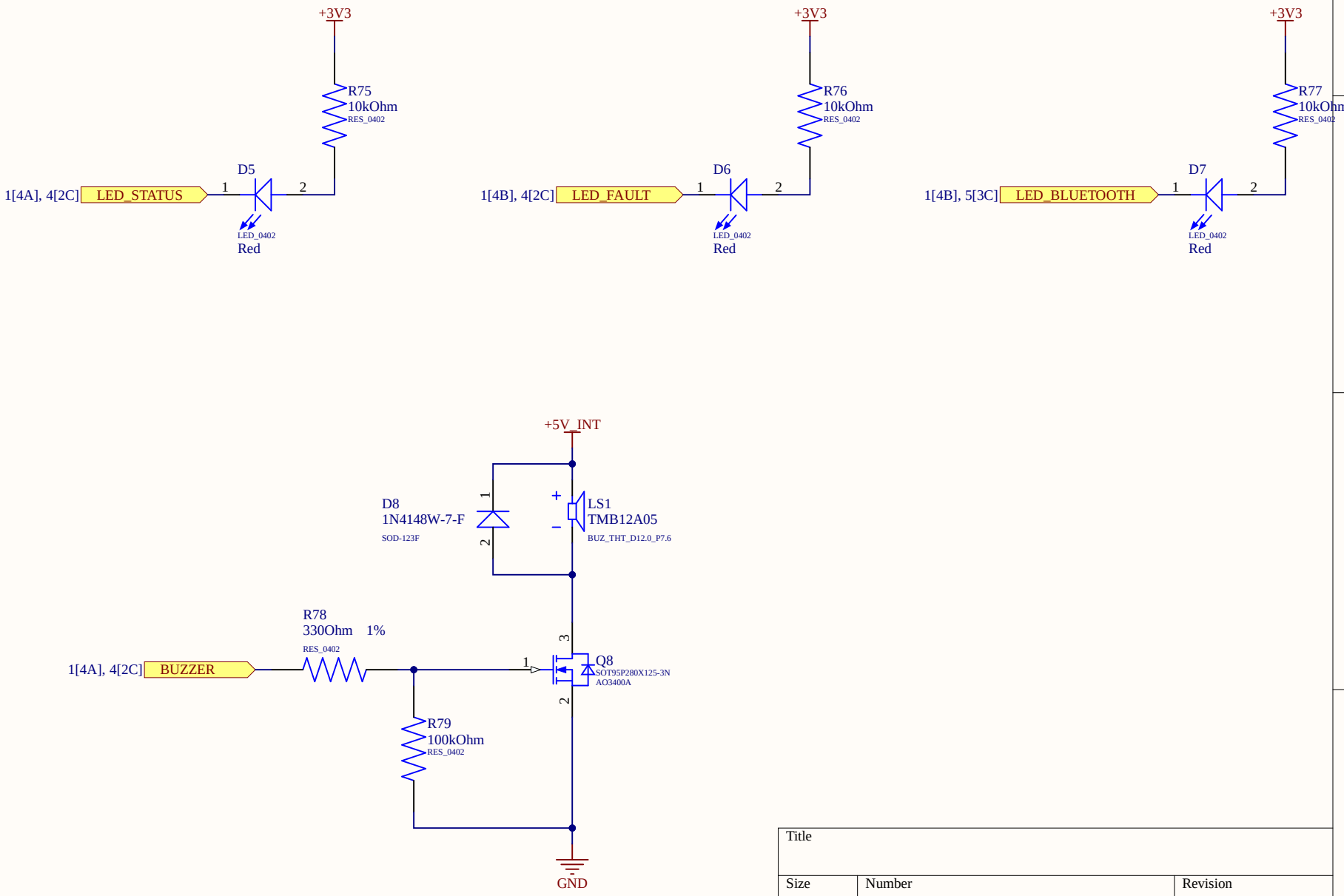


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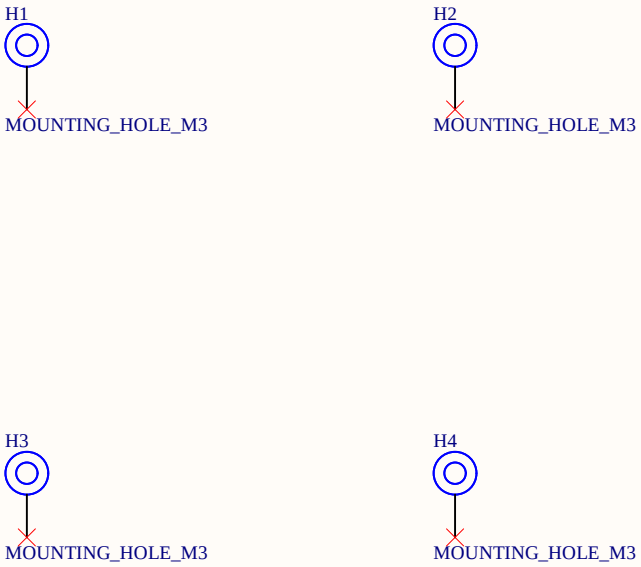

+5V_EXT
C77
100nF
50V
N_P_CAP_0603
GND
PE_RX_EXT
PE_TX_EXT
J30
1
2
3
4
CONN_HY2.0_1x04_P2.00mm_TH_RA
U19
PE_TX_EXT
PE_RX_EXT
+5V_EXT
GND
USBLC6-2SC6
+3V3
SB3
SOLDERING_BRIDGE_3P
R67
10kOhm
RES_0402
R68
10kOhm
RES_0402
R69
330Ohm 1%
RES_0402
1[4D], 5[3C]
PE_TX
PE_TX_EXT
Q6
BSS138LT1G
500mV
SOT95P280X125-3N
SB4
R70
10kOhm
RES_0402
R71
10kOhm
RES_0402
R72
330Ohm 1%
RES_0402
PE_RX
PE_RX_EXT
Q7
BSS138LT1G
500mV
SOT95P280X125-3N
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Number
Revision
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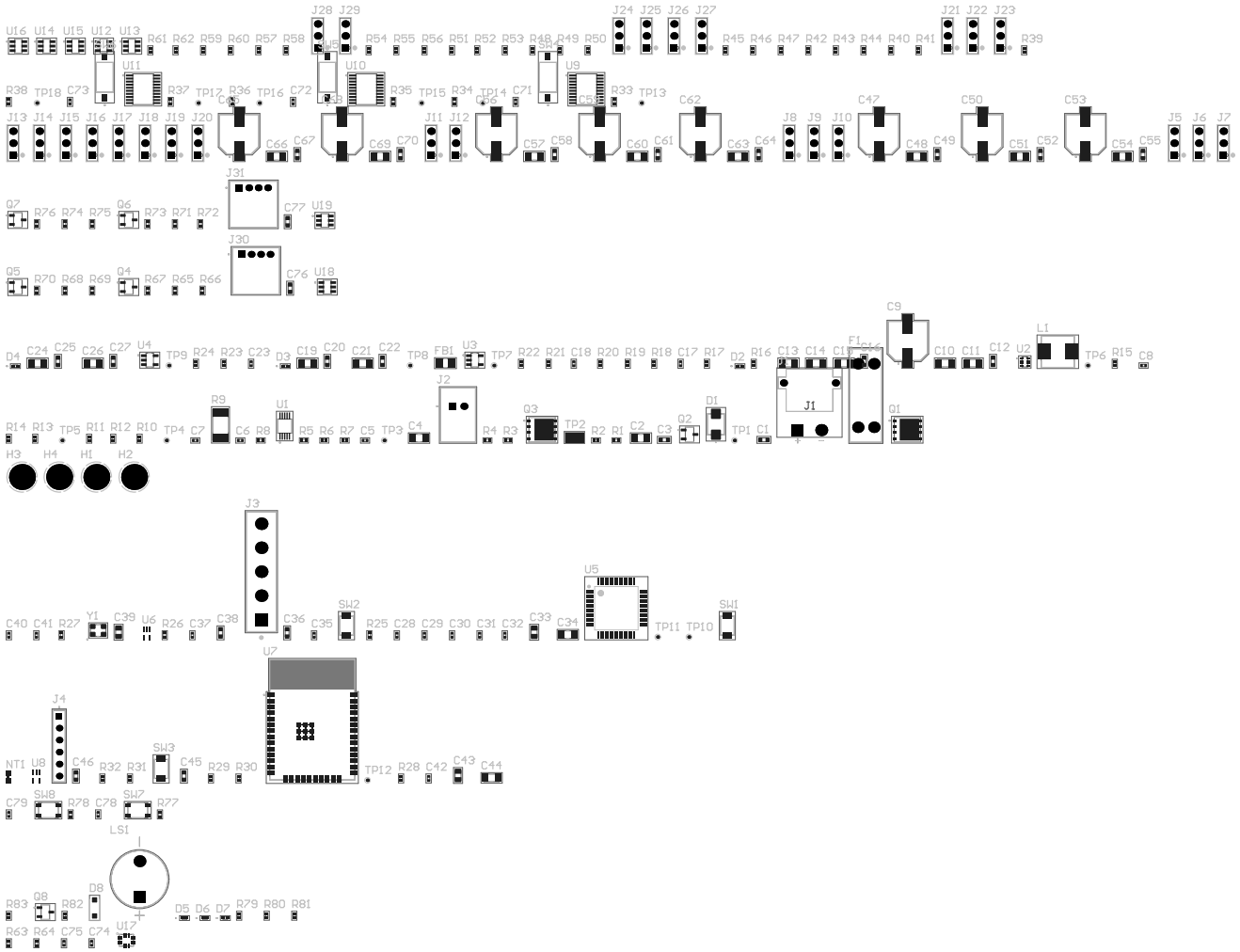
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U16 U14 U15 U12 U13 R61 R62 R59 R60 R57 R58 J28 J29 R54 R55 R56 R51 R52 R53 R48 R49 R50 J24 J25 J26 J27 R45 R46 R47 R42 R43 R44 R40 R41 J21 J22 J23 R39
U11 U10 U9
R38 TP18 C73 R37 TP17 R36 TP16 C72 R35 TP15 R34 TP14 C71 R33 TP13
C62
J13 J14 J15 J16 J17 J18 J19 J20 C66 C67 C69 C70 J11 J12 C57 C58 C60 C61 C63 C64 J8 J9 J10 C47 C48 C49 C50 C51 C52 C53 C54 C55 J5 J6 J7
J31
Q7 R76 R74 R75 Q6 R73 R71 R72 C77 U19
J30
Q5 R70 R68 R69 Q4 R67 R65 R66 C76 U18
C9
D4 C24 C25 C26 C27 U4 TP9 R24 R23 C23 D3 C19 C20 C21 C22 TP8 FB1 U3 TP7 R22 R21 C18 R20 R19 R18 C17 R17 D2 R16 C13 C14 C15 F1 C10 C11 C12 U2 L1 TP6 R15 C8
J1
R14 R13 TP5 R11 R12 R10 TP4 C7 R9 C6 R8 U1 R5 R6 R7 C5 TP3 C4 J2 R4 R3 Q3 TP2 R2 R1 C2 C3 Q2 D1 TP1 C1 Q1
H3 H4 H1 H2
J3
C40 C41 R27 Y1 C39 U6 R26 C37 C38 C36 C35 SW2 R25 C28 C29 C30 C31 C32 C33 C34 U5 TP11 TP10 SW1
U7
J4
NT1 U8 C46 R32 R31 SW3 C45 R29 R30 TP12 R28 C42 C43 C44
C79 SW8 R78 C78 SW7 R77
LS1
R83 Q8 R82 D8 D5 D6 D7 R79 R80 R81
R63 R64 C75 C74 U17